

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 25

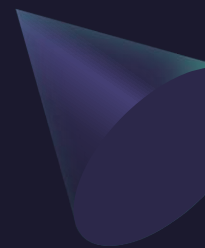


Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 16.5, 16.6 (curl, divergence, parametric surfaces)
- Exam 3: Friday May 1 – Ch 13.1-13.3 and 16.1-16.8
- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104



ALVARO: Start the recording!

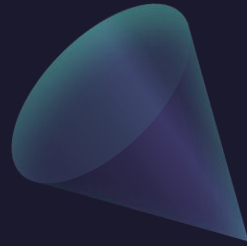
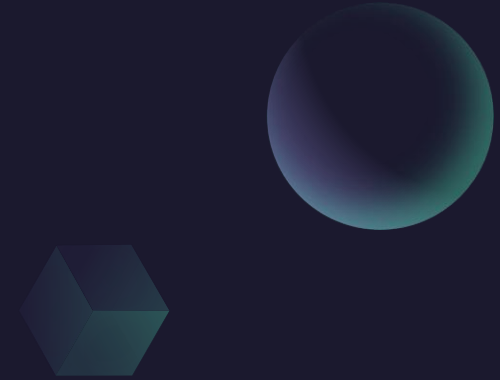


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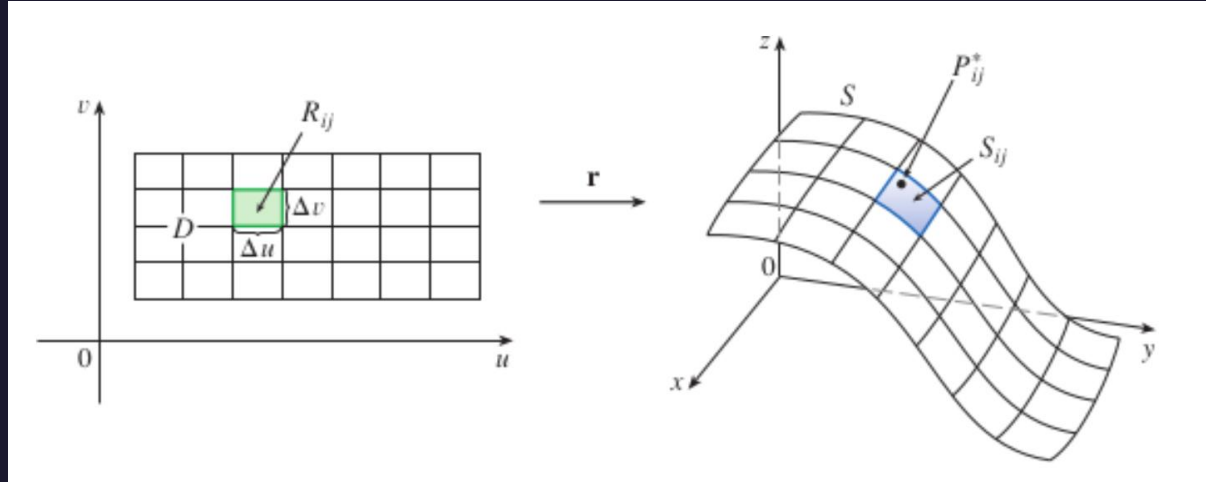
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Surface Integrals



Integrals on Parametrized Surfaces

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$



$$\iint_S f(x, y, z) \, dS = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(P_{ij}^*) \, \Delta S_{ij}$$

$$\iint_S f(x, y, z) \, dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| \, dA$$

Integrals on Graphs of Functions

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$

$$\iint_S f(x, y, z) dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| dA$$

$$\iint_S f(x, y, z) dS = \iint_D f(x, y, g(x, y)) \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} dA$$

Orientation

- ▶ For surface integrals of a vector field $\vec{F}$ over a surface S , the orientation of the surface must be specified, just like the line integrals. The orientation of a surface is specified via its normal vectors at any point.
- ▶ For a closed surface S , we define **the positive orientation** of S to mean that all normal vectors to S point outward (with negative meaning normal vectors point inward instead). If it is impossible to have all normal vectors always point either outward or inward, then the surface S is non-orientable.

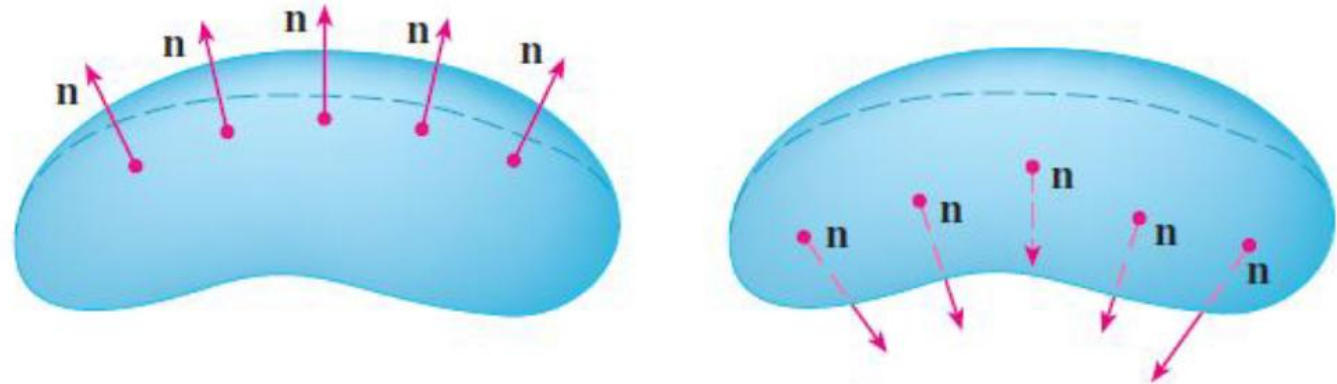


FIGURE 7

The two orientations
of an orientable surface

- ▶ To define a surface integral of a vector field, we only want to count the portion of $\vec{F}$ that is normal to the surface S at any point. Hence, the dot product again plays a role in the computation.
- ▶ Then the **surface integral** of $\vec{F}$ over the surface S is defined as

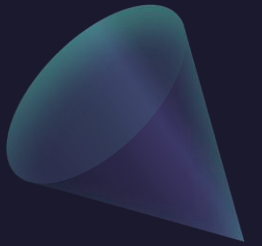
$$\begin{aligned}\iint_S \vec{F} \cdot d\vec{S} &= \iint_S \vec{F} \cdot \vec{n} \, dS \\ &= \iint_S \vec{F} \cdot \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} \, dS \\ &= \iint_D \left[\vec{F} \cdot \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} \right] |\vec{r}_u \times \vec{r}_v| \, dA \\ &= \iint_D \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) \, dA.\end{aligned}$$

This is often called the **flux integral** as it measures the flux of $\vec{F}$ across the surface S .

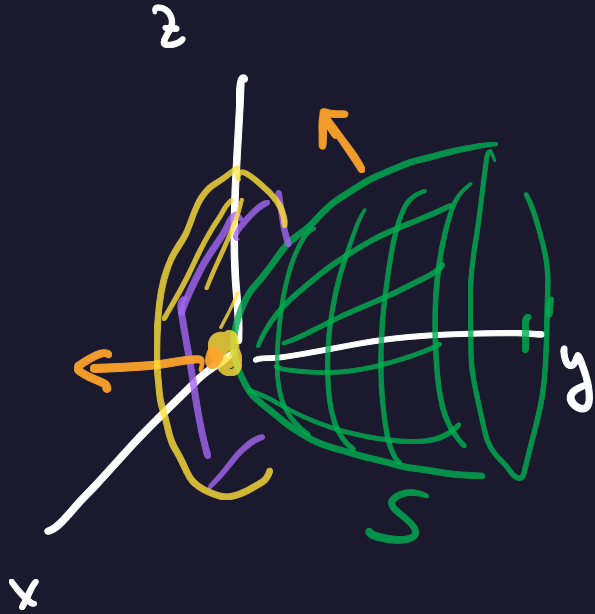
- ▶ Recall and compare it with the line integral of vector fields:

$$\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) \, dt.$$

Orientation



Example: Find the flux of the vector field $F = (0, y, -z)$ across the surface S given by $y = x^2 + z^2$, $0 \leq y \leq 1$.
(Assume S is positively oriented.)



$$\iint_S F \cdot d\mathbf{S} \quad \begin{cases} \mathbf{r}(x,z) = (x, x^2+z^2, z) \\ \mathbf{r}_x = (1, 2x, 0) \\ \mathbf{r}_z = (0, 2z, 1) \end{cases}$$

$$\mathbf{r}_x \times \mathbf{r}_z = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2x & 0 \\ 0 & 2z & 1 \end{vmatrix} = (2x, -1, 2z)$$

$$\mathbf{r}_x \times \mathbf{r}_z \Big|_{(0,0)} = (0, -1, 0) \quad D = \{(x,z), x^2+z^2 \leq 1\}$$

$$F(\mathbf{r}(x,z)) = (0, x^2+z^2, -z)$$

Example: Find the flux of the vector field $F = (0, y, -z)$ across the surface S given by $y = x^2 + z^2$, $0 \leq y \leq 1$.
(Assume S is positively oriented.)

$$\iint_S F \cdot dS = \iint_D (-x^2 - 3z^2) dA \quad D = \{(x, z) : x^2 + z^2 \leq 1\}$$

$$\left. \begin{aligned} F(r(x, z)) &= (0, x^2 + z^2, -z) \\ \frac{1}{2} r_x \times r_z &= (2x, -1, 2z) \end{aligned} \right\} \begin{aligned} \text{DOT} &= 0 + (-1)(x^2 + z^2) - 2z^2 \\ &= -x^2 - 3z^2 \end{aligned}$$

$$F(r(x, z)) \cdot (r_x \times r_z)$$

Example: Find the flux of the vector field $F = (0, y, -z)$ across the surface S given by $y = x^2 + z^2$, $0 \leq y \leq 1$.
(Assume S is positively oriented.)

$$\iint_D (-x^2 - 3z^2) dA = -\iint_D x^2 dA - 3 \iint_D z^2 dA$$

$$\iint_D x^2 dA = \iint_D z^2 dA \quad D = \{x^2 + z^2 \leq 1\}$$

Polar
coords

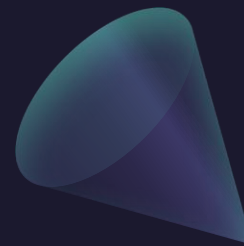
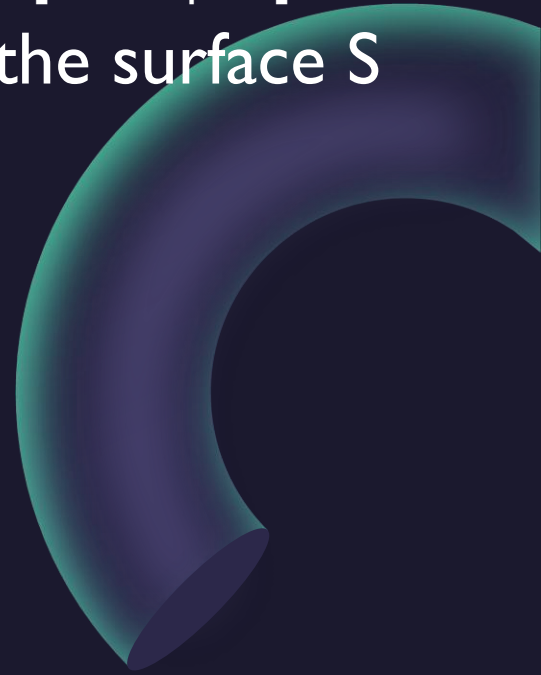
$\parallel$
 $\pi/4$

$\pi/4$

$$= -\pi/4 - 3\pi/4 = \boxed{-\pi}$$

...

Example: Find the flux of the vector field $F = (0, y, -z)$ across the surface S given by $y = x^2 + z^2$, $0 \leq y \leq 1$. (Assume S is positively oriented.)



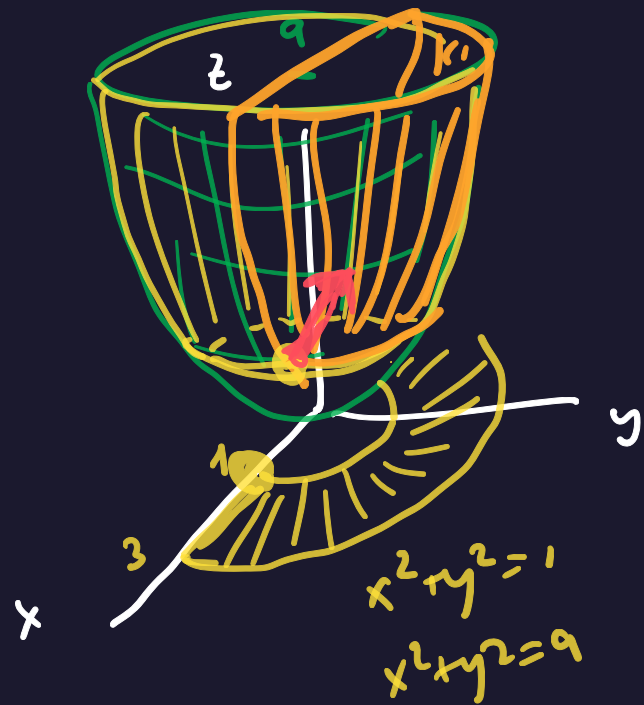
Example: Let F be the vector field $F = (z^2, x^2, y^2)$, and the surface S be given by $z = x^2 + y^2$, $1 \leq z \leq 9$, $y \geq 0$. Assume S is positively oriented. Set up (but do not evaluate).

$$\iint_S \text{curl}(F) \cdot dS$$

$$F = (z^2, x^2, y^2), \quad \nabla \times F = \text{Curl}(F)$$

$$= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z^2 & x^2 & y^2 \end{vmatrix} = \dots = (2y, 2z, 2x)$$

$$i \begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & y^2 \end{vmatrix} = i \left(\frac{\partial}{\partial y} y^2 - \frac{\partial}{\partial z} x^2 \right) = i(2y - 0) = i(2y)$$



Example: Let F be the vector field $F = (z^2, x^2, y^2)$, and the surface S be given by $z = x^2 + y^2$, $1 \leq z \leq 9$, $y \geq 0$. Assume S is positively oriented. Set up (but do not evaluate).

$$\iint_S \text{curl}(F) \cdot dS$$

$$D = \{(x, y) : 1 \leq x^2 + y^2 \leq 9, y \geq 0\}$$

$$r(x, y) = (x, y, x^2 + y^2)$$

$$r_x = (1, 0, 2x)$$

$$r_y = (0, 1, 2y)$$

$$r_x \times r_y = \begin{vmatrix} i & j & k \\ 1 & 0 & 2x \\ 0 & 1 & 2y \end{vmatrix} = (-2x, -2y, 1)$$

$$= i|4| - j|2| + k|2|$$

$$\begin{pmatrix} 1, 0 \\ -2, 0, 1 \end{pmatrix} = n$$

Example: Let F be the vector field $F = (z^2, x^2, y^2)$, and the surface S be given by $z = x^2 + y^2$, $1 \leq z \leq 9$, $y \geq 0$.

Assume S is positively oriented. Set up (but do not evaluate).

$$r(x, y) = (x, y, x^2 + y^2)$$

$$\iint_S \text{curl}(F) \cdot dS$$

$$\iint_S G \cdot dS$$

$$= \iint_D G(r(u, v)) \cdot (r_u \times r_v) \, dA$$

$$\text{Curl}(F) = \nabla \times F = (2y, 2z, 2x)$$

$$\text{Curl}(F)(r(x, y)) = (2y, 2(x^2 + y^2), 2x)$$

$$\text{Curl}(F)(r(x, y)) \cdot (r_x \times r_y) = (2y, 2(x^2 + y^2), 2x) \cdot (-2x, -2y, 1)$$

$$= -4xy - 4y(x^2 + y^2) + 2x$$

Questions?

$$D = \{(x, y) : 1 \leq x^2 + y^2 \leq 9, y \geq 0\}$$

$$\iint_S \text{Curl}(F) \cdot dS$$

$$= \iint_D (-4xy - 4y(x^2 + y^2) + 2x) dA$$

$$D = \{(r, \theta) : 1 \leq r \leq 3, 0 \leq \theta \leq \pi\}$$

polar?

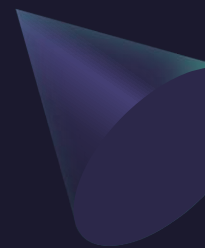
$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$x^2 + y^2 = r^2 \dots$$



ALVARO: Start the recording!



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Stokes' Theorem



Recall: Green's Theorem

Green's Theorem

Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C . If P and Q have continuous partial derivatives on an open region that contains D , then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

$\mathbf{F} = (P, Q)$



Vector Form of Green's Theorem



Consider $\mathbf{F} = (P, Q)$ satisfying Green's Theorem hypotheses.

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C P dx + Q dy$$

$$(\text{curl } \mathbf{F}) \cdot \mathbf{k} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \mathbf{k} \cdot \mathbf{k} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

Then Green's Theorem says:

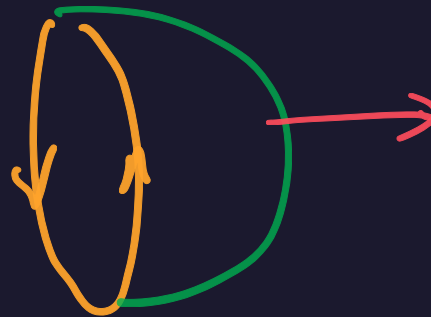
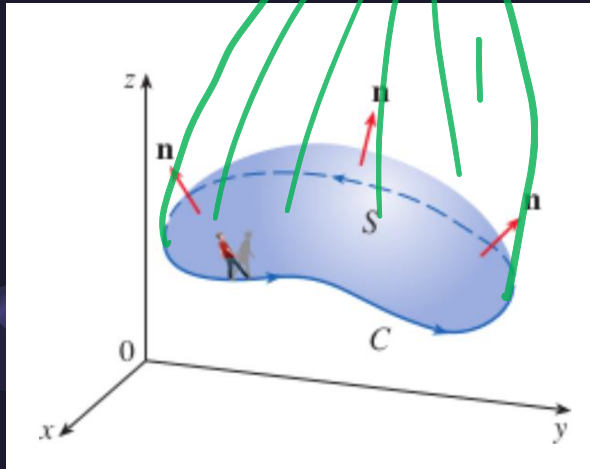
$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C \mathbf{F} \cdot \mathbf{T} ds = \iint_D (\text{curl } \mathbf{F}) \cdot \mathbf{k} dA$$

Stokes' Theorem

Stokes' Theorem

Let S be an oriented piecewise-smooth surface that is bounded by a simple, closed, piecewise-smooth boundary curve C with positive orientation. Let $\mathbf{F}$ be a vector field whose components have continuous partial derivatives on an open region in $\mathbb{R}^3$ that contains S . Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$$



Sir
George Stokes
FRS



Stokes, 1860s

35th **President of the Royal Society**

In office
1885–1890

Preceded by [Thomas Huxley](#)

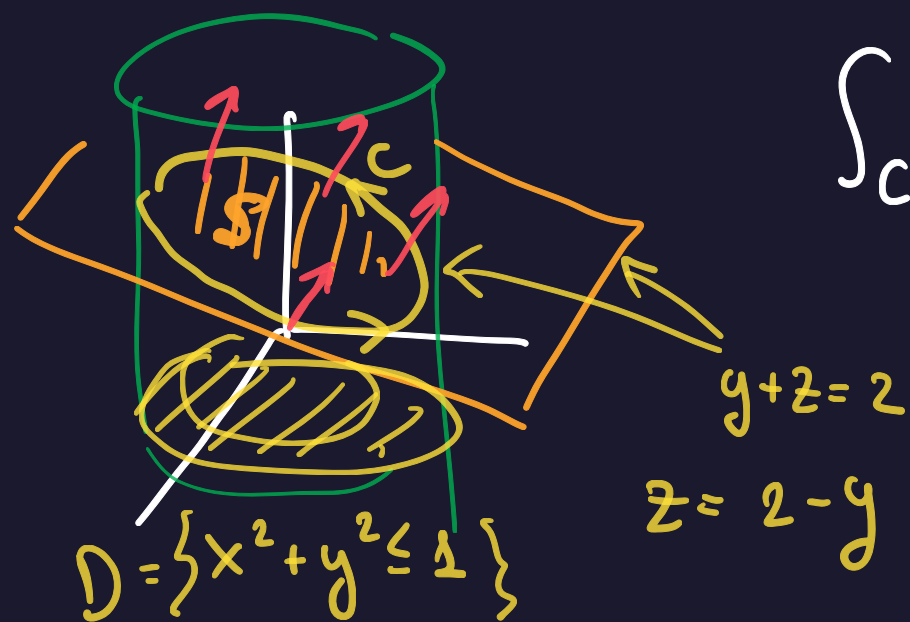
Succeeded by [Lord Kelvin](#)

Personal details

Born George Gabriel Stokes
13 August 1819
[Skreen](#), Ireland, United Kingdom of Great Britain and Ireland

Died 1 February 1903 (aged 83)
[Cambridge](#), England, United Kingdom of Great Britain and Ireland

Example: Evaluate $\int_C F \cdot dr$ where $F = (-y^2, x, z^2)$ and C is the curve of intersection of the plane $y + z = 2$ and the cylinder $x^2 + y^2 = 1$. (C is oriented counterclockwise when viewed from above.)



$$r(x, y) = (x, y, 2 - y)$$

$$\int_C F \cdot dr = \iint_S \text{Curl}(F) \cdot dS$$

$$r_x = (1, 0, 0)$$

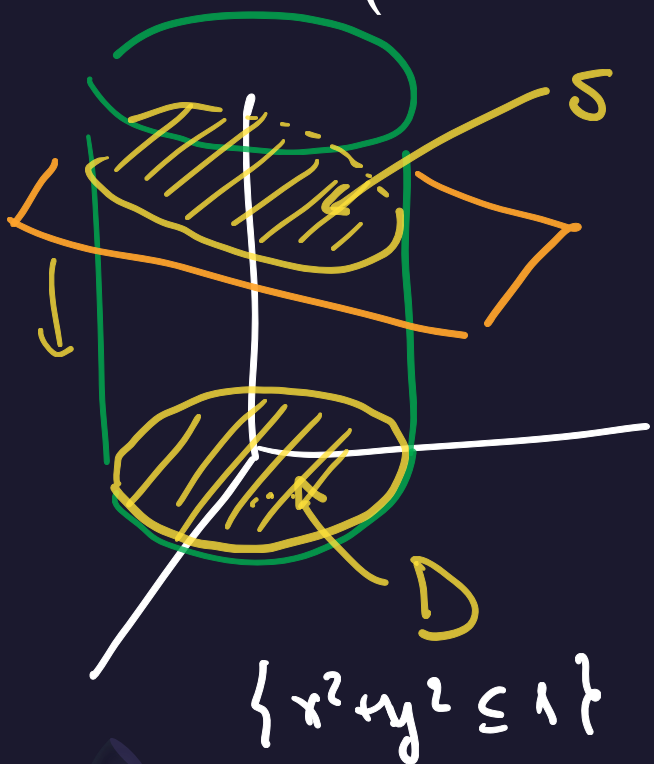
$$r_y = (0, 1, -1)$$

$$r_x \times r_y = \begin{vmatrix} i & j & k \\ 1 & 0 & 0 \\ 0 & 1 & -1 \end{vmatrix} = (0, 1, 1) = n$$

pos. oriented!

STOKES!

Example: Evaluate $\int_C F \cdot dr$ where $F = (-y^2, x, z^2)$ and C is the curve of intersection of the plane $y + z = 2$ and the cylinder $x^2 + y^2 = 1$. (C is oriented counterclockwise when viewed from above.)



$$\text{Curl}(F) = \nabla \times F$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y^2 & x & z^2 \end{vmatrix} = (0, 0, 1 + 2y)$$

$$r(x, y) = (x, y, 2 - y)$$

$$\nabla \times F(r(x, y)) = (0, 0, 1 + 2y)$$

$$\nabla \times F(r) \cdot (0, 1, 1) = 1 + 2y$$

$$\int_C F \cdot dr$$

$$\iint_S \nabla \times F \cdot d\mathbf{S}$$

$$= \iint_D (1 + 2y) dA$$

$$= \iint_D 1 dA + 2 \iint_D y dA$$

$$\pi$$

Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (xz, yz, xy)$ and S is the part of the sphere of radius 2 centered at the origin that is inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane.



$$\boxed{0} = \iint_S \text{curl}(F) dS = \int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t) dt$$

$$\left. \begin{array}{l} x^2 + y^2 + z^2 = 4 \\ x^2 + y^2 = 1 \end{array} \right\} \begin{array}{l} 1 + z^2 = 4 \Rightarrow z^2 = 3 \Rightarrow z = \sqrt{3} \end{array}$$

$$r(t) = (\cos t, \sin t, \sqrt{3})$$

$$r'(t) = (-\sin t, \cos t, 0)$$

$$F(r(t)) = (\sqrt{3} \cos t, \sqrt{3} \sin t, \cos t \sin t)$$

$$0 \leq t \leq 2\pi \quad F(r) \cdot r'$$

$$\left. \begin{array}{l} -\sqrt{3} \cos t \sin t \\ +\sqrt{3} \cos t \sin t \end{array} \right\} \int_0^{2\pi} dt = 0$$

Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (xz, yz, xy)$ and S is the part of the sphere of radius 2 centered at the origin that is inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane.

$$\text{curl}(F) = (x-y, x-y, 0)$$

$$z^2 = 1 - x^2 - y^2$$

$$z = \sqrt{1 - x^2 - y^2}$$

$$r(x, y) = (x, y, \sqrt{1 - x^2 - y^2})$$

$$r_x =$$

$$r_y =$$

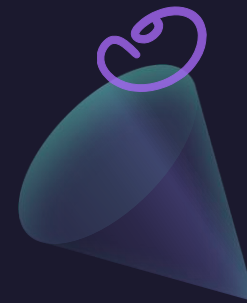
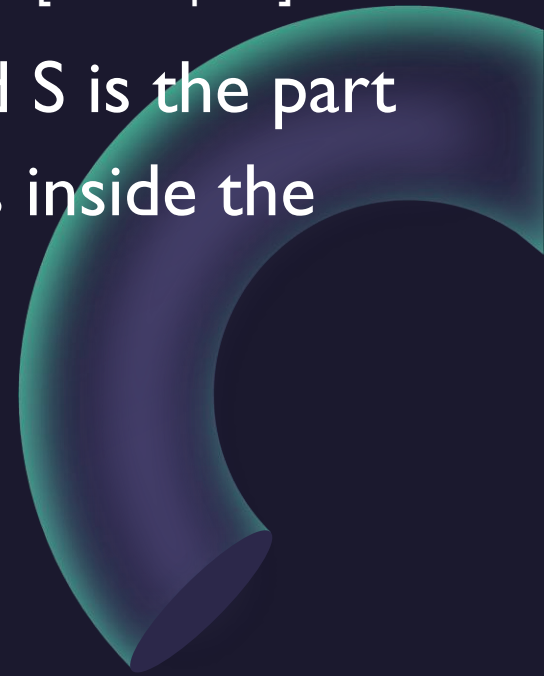
$$r_x \times r_y = \dots$$

Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (xz, yz, xy)$ and S is the part of the sphere of radius 2 centered at the origin that is inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane.

Alternative solution: Change S to a disk!

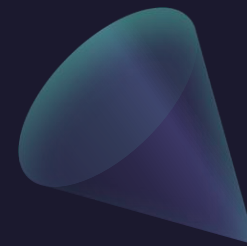
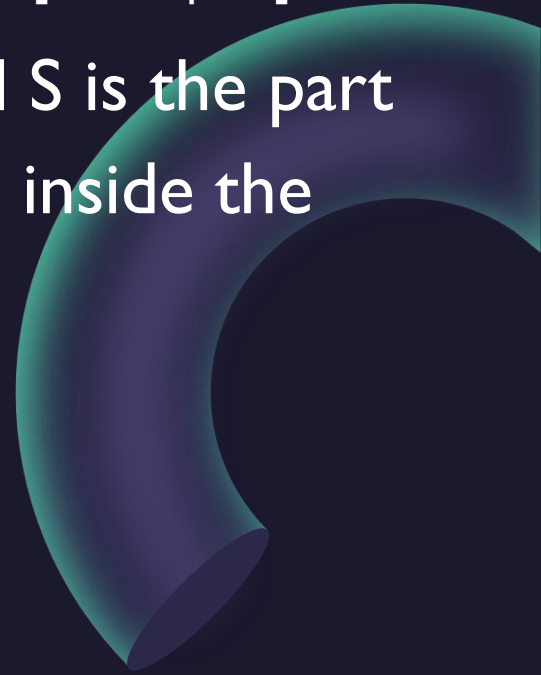


STOKES

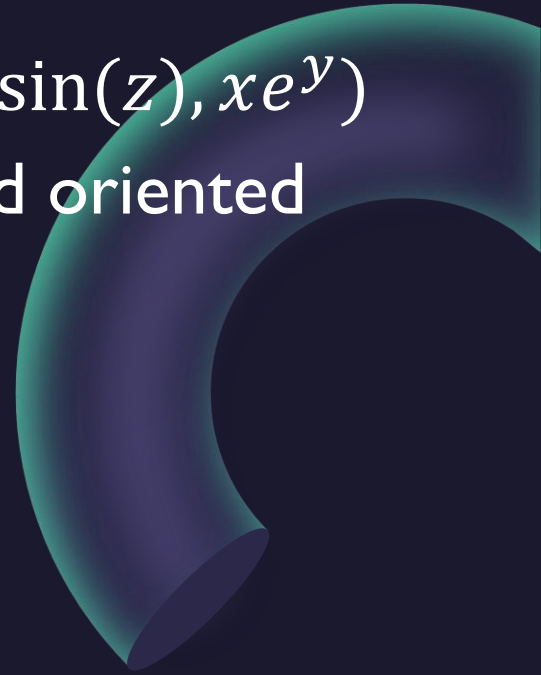
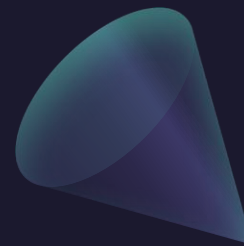


Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (xz, yz, xy)$ and S is the part of the sphere of radius 2 centered at the origin that is inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane.

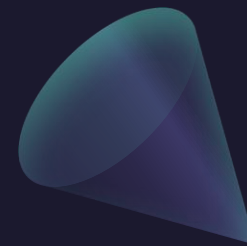
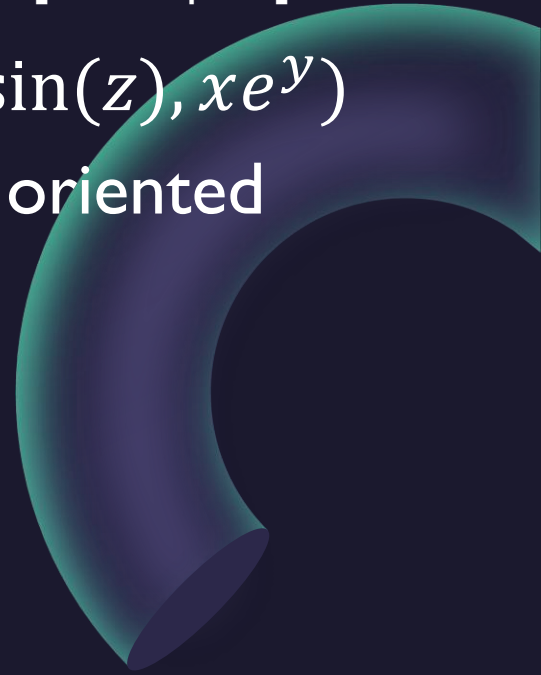
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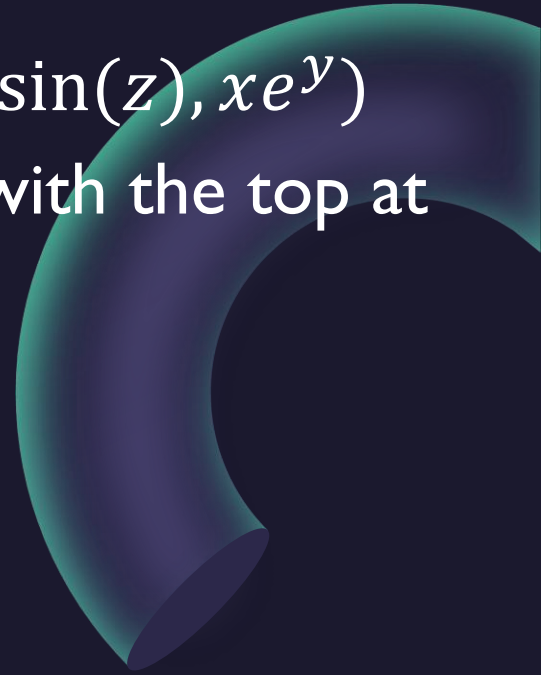
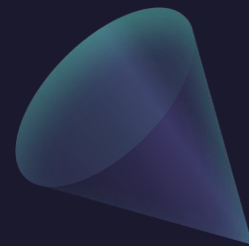
Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (2y \cos(z), e^x \sin(z), xe^y)$ and S is the semisphere $x^2 + y^2 + z^2 = 9$, $z \geq 0$ and oriented upwards.



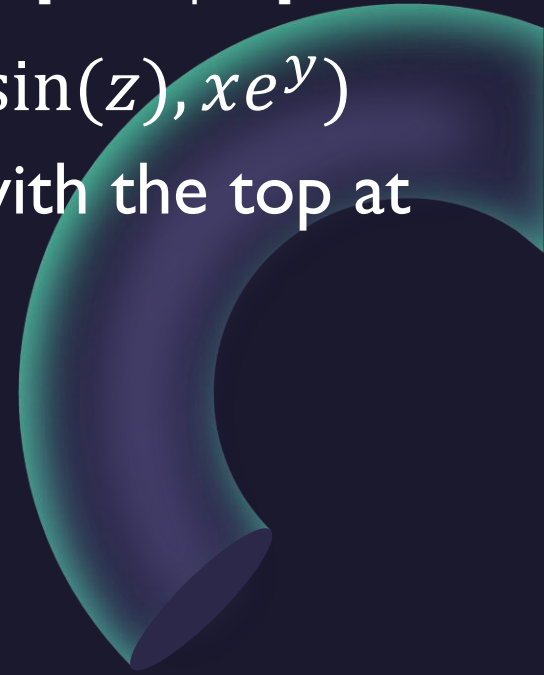
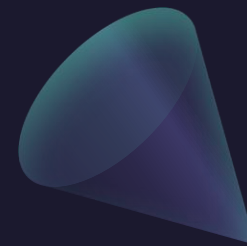
Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (2y \cos(z), e^x \sin(z), xe^y)$ and S is the semisphere $x^2 + y^2 + z^2 = 9$, $z \geq 0$ and oriented upwards.



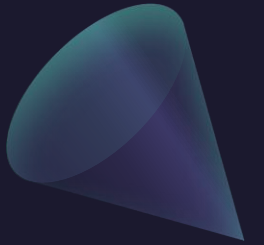
Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (2y \cos(z), e^x \sin(z), xe^y)$ and S is the cylinder $x^2 + y^2 = 9$, $0 \leq z \leq 4$ along with the top at $z = 4$.



Example: Evaluate $\iint_S \text{curl}(F) \cdot dS$ where $F = (2y \cos(z), e^x \sin(z), xe^y)$ and S is the cylinder $x^2 + y^2 = 9$, $0 \leq z \leq 4$ along with the top at $z = 4$.

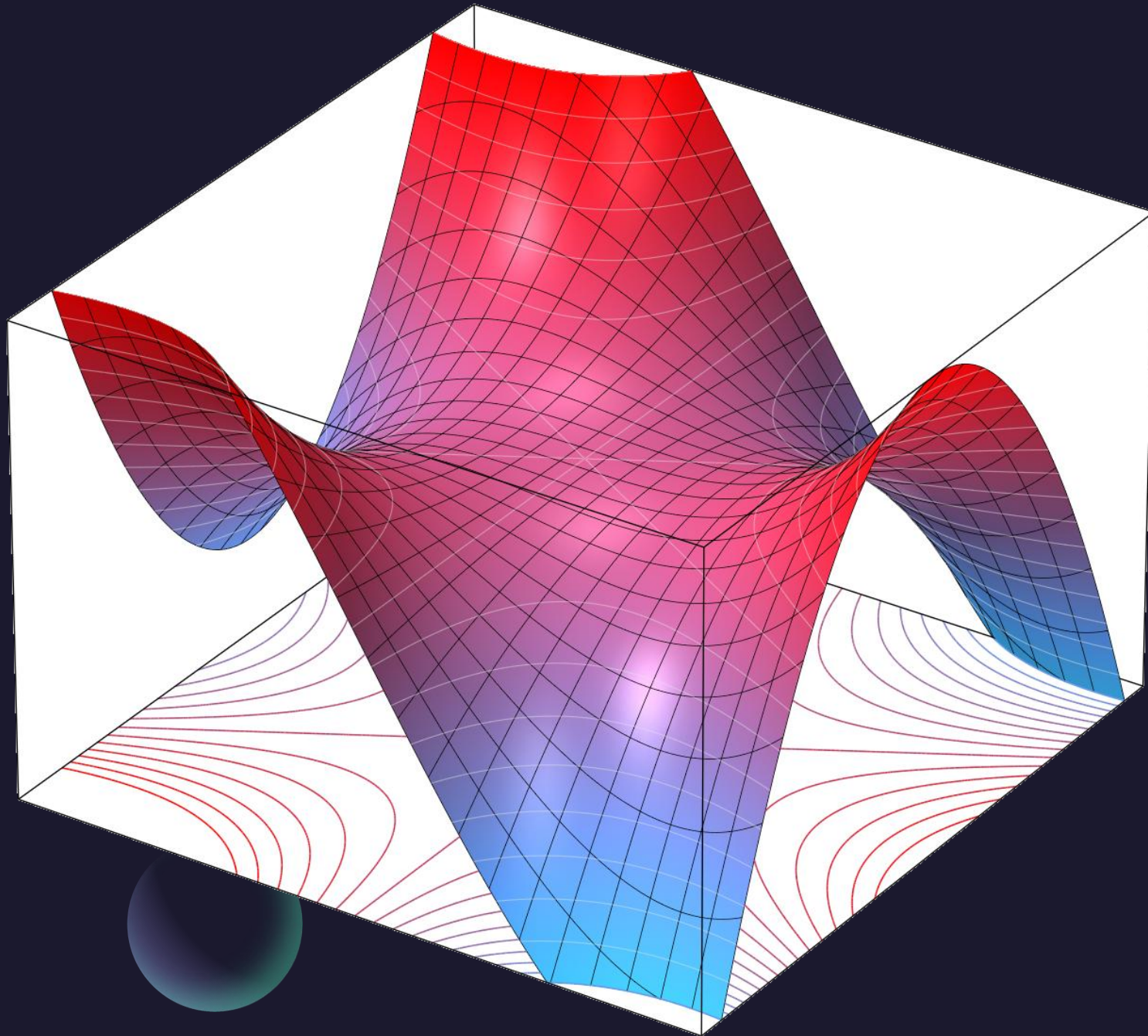


Questions?



Thank you

Until next time.



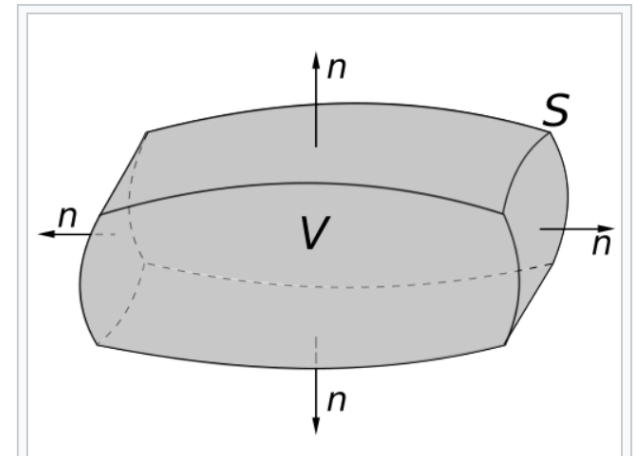
The Divergence Theorem (Gauss's Theorem)

Mathematical statement [\[edit\]](#)

Suppose V is a [subset](#) of $\mathbb{R}^n$ (in the case of $n = 3$, V represents a volume in [three-dimensional space](#)) which is [compact](#) and has a [piecewise smooth boundary](#) S (also indicated with $\partial V = S$). If $\mathbf{F}$ is a continuously differentiable vector field defined on a [neighborhood](#) of V , then:[\[4\]](#)[\[5\]](#)

$$\iiint_V (\nabla \cdot \mathbf{F}) \, dV = \oiint_S (\mathbf{F} \cdot \hat{\mathbf{n}}) \, dS.$$

The left side is a [volume integral](#) over the volume V , and the right side is the [surface integral](#) over the boundary of the volume V . The closed, measurable set ∂V is oriented by outward-pointing [normals](#), and $\hat{\mathbf{n}}$ is the outward pointing unit normal at almost each point on the boundary ∂V . ($d\mathbf{S}$ may be used as a shorthand for $\mathbf{n}dS$.) In terms of the intuitive description above, the left-hand side of the equation represents the total of the sources in the volume V , and the right-hand side represents the total flow across the boundary S .



A region V bounded by the surface $S = \partial V$ with the surface normal n